

Multi-Agent Prompting

Multi-Agent Prompting 7-Day Vegetarian South Indian Weight-Loss Meal Planner

Goal: Use several specialized roles to evaluate the meal plan from different perspectives and combine their contributions.

User Profile

Age: 35 years | Weight: 75 kg | Height: 170 cm

Activity: Moderately Active | Cooking time: 30 minutes | People: 2

Restrictions: None | Location: Chennai / Tamil Nadu | Budget: Moderate

Prompt Example

Act as a coordinated team of specialized AI agents working on one task: create a **7-day vegetarian South Indian weight-loss meal plan** for the user profile above.

Use these roles:

Agent 1 - Nutrition Planner: Focus on balanced meals, vegetarian protein, vegetables, whole grains, sensible portions, and sustainable weight-loss principles.

Agent 2 - South Indian Food Specialist: Ensure meals are culturally appropriate, familiar, locally available, and realistic for Chennai / Tamil Nadu.

Agent 3 - Practicality & Budget Reviewer: Check the 30-minute weekday cooking limit, ingredient reuse, affordability, batch preparation, and waste reduction.

Agent 4 - Quality Reviewer: Review the combined plan for repetition, balance, missing meal categories, impractical choices, and consistency.

Have the roles contribute their recommendations, then synthesize them into one coherent final plan. Do not expose private hidden reasoning. Provide only concise role-level findings and the integrated result.

Required Output

1. Short findings from each specialist role
2. Integrated 7-day meal-plan table
3. Weekly grocery list
4. Preparation, portion-control, and substitution tips
5. Final quality-check summary

How Multi-Agent Works

Nutrition Agent + Cuisine Agent + Practicality Agent -> Quality Reviewer -> Integrated Final Plan

Multi-agent prompting assigns different perspectives to specialized roles. Their contributions are then synthesized and reviewed, helping a complex task benefit from nutritional, cultural, practical, and

quality-control perspectives rather than relying on a single viewpoint.

Note: This is a prompt-engineering example, not individualized dietary or medical advice. For personalized nutrition guidance, consult a qualified dietitian.